

SyllabusSync AI — Deep Research & Hackathon Strategy

Core Idea

Build an AI-powered academic planning assistant that converts messy course syllabi, CourseLink announcements, emails, and assignment descriptions into:

- A unified academic calendar
- Smart weekly task planning
- AI-generated study breakdowns
- Risk/overload detection
- Auto-generated reminders and todos
- Exportable calendar events (.ics / Google Calendar)

The goal is to solve a real student problem:

"I missed an assignment because the syllabus was 18 pages long and hidden inside random PDFs and announcements."

This is strong hackathon material because:

- Extremely relatable
- Easy to demo visually
- Uses AI meaningfully
- Feels like a real startup product
- Can be completed in 24 hours with MVP scope
- Lets you show OCR, LLMs, structured extraction, and automation

Why This Idea Is Strong

1. Immediate Understandability

Judges understand the value in under 10 seconds.

"Upload your syllabi → instantly get your semester planned out."

That is MUCH stronger than:

- generic chatbot
- AI study buddy

- random Gemini wrapper
- motivational AI

2. High Visual Impact

You can create an impressive demo:

Demo Flow

1. Drag in 2–3 real syllabi PDFs
2. Show messy raw PDF text
3. AI extracts:
 4. assignments
 5. quizzes
 6. midterms
 7. weights
 8. dates
 9. lecture topics
10. Timeline/calendar auto-generates
11. AI warns:
 12. "Week 7 overload detected"
 13. "3 midterms within 5 days"
14. Show generated study tasks
15. Export to Google Calendar
16. Show notifications/reminders

Judges LOVE visible transformation.

3. You Can Add Layers Fast

Even if the MVP is simple, you can stack features quickly:

Basic MVP

- PDF upload
- date extraction
- calendar output

Advanced Version

- AI prioritization
- burnout detection

- study time estimation
- natural-language planner
- assignment decomposition
- chatbot assistant
- collaborative planning
- professor announcement ingestion

This gives you infinite stretch-goal potential.

The Main Product Vision

Elevator Pitch

"SyllabusSync AI turns chaotic university course documents into an intelligent semester planner that automatically schedules deadlines, generates study plans, detects overload weeks, and helps students stay ahead before they fall behind."

Real Problems Students Have

Pain Points

1. Syllabi Are Massive

Students receive:

- 10–25 page PDFs
- weird formatting
- inconsistent dates
- hidden grading rules

2. Important Dates Are Everywhere

Deadlines exist in:

- syllabi
- announcements
- emails
- assignment docs
- CourseLink
- Discord messages

3. Students Only Realize They're Overloaded Too Late

Example:

- midterm Monday
- assignment Tuesday
- lab Wednesday
- quiz Friday

Nobody notices until the week arrives.

4. Existing Calendar Apps Are Manual

Google Calendar requires manual entry. Students never fully maintain it.

Your app automates everything.

Feature Research

Core Features (MVP)

1. PDF Upload + Parsing

Input

- syllabus PDFs
- DOCX
- pasted text
- screenshots (optional)

Output

Structured JSON:

```
{
  "course": "CIS1910",
  "assignments": [
    {
      "title": "Assignment 1",
      "due_date": "2026-09-20",
      "weight": "10%"
    }
  ]
}
```

```
]
}
```

Recommended Stack

Frontend:

- React
- Next.js
- Tailwind

Backend:

- FastAPI OR Node.js Express

PDF Parsing:

- pdfplumber
- PyMuPDF
- unstructured
- LangChain loaders

OCR:

- Tesseract
- Gemini Vision

2. AI Date & Assignment Extraction

This is the most important feature.

The AI should extract:

- assignment names
- due dates
- exam dates
- grade weights
- lecture topics
- office hours
- submission platforms

Recommended AI Approach

Use structured extraction prompts.

Example prompt:

Extract all assignments, exams, quizzes, and deadlines from this syllabus.
Return ONLY valid JSON.

Best Models

Best hackathon choices:

- Gemini 2.5 Flash
- Gemini 2.5 Pro
- GPT-4.1 Mini
- Claude Sonnet

Gemini is especially good because:

- PDF support
 - multimodal
 - generous free credits at hackathons
 - strong structured extraction
-

3. Unified Semester Calendar

Calendar Views

- monthly
- weekly
- timeline
- heatmap

Libraries

- FullCalendar
- React Big Calendar
- Tremor

Important Feature

Color-code by course.

Example:

- CIS1910 = blue
- STAT2040 = green
- BIOL1090 = red

This massively improves visual clarity.

4. Smart Weekly Planner

This is where the project becomes more than a parser.

The AI should generate:

This week:

- Finish CIS1910 Assignment 2 draft
- Review recursion for quiz
- Begin STAT2040 practice questions
- Spend 3 hours preparing lab report

Advanced Version

Estimate workload:

Expected workload this week: 16 hours
Risk level: HIGH

5. Overload / Burnout Detection

One of the strongest features.

Example

Week 8 detected as high intensity.

You have:

- 2 midterms
- 1 lab
- 1 essay

Suggested action:

Begin studying 10 days earlier.

Why Judges Like This

It feels intelligent. Not just "AI summary app #3847".

6. Calendar Export

Export Types

- .ics
- Google Calendar
- Outlook

Easy Win

This gives immediate real-world usefulness.

Libraries

- ical-generator
 - ics npm package
-

Advanced Features (Stretch Goals)

1. CourseLink / LMS Integration

Potential integrations:

- Canvas
- Moodle
- CourseLink
- Brightspace

Even mock integration screenshots are enough for hackathons.

2. AI Assignment Breakdown

Example:

Input: "Research paper due in 3 weeks"

AI Output:

```
Week 1:  
- choose topic  
- gather sources
```


Week 2:

- write outline
- draft introduction

Week 3:

- finalize citations
- proofreading

This makes the app feel proactive.

3. Natural Language Chat

Examples:

"What should I focus on this weekend?"
"When is my busiest month?"
"What assignments are worth the most?"
"What can I safely delay?"

4. Study Time Estimation

AI estimates required effort.

Example:

Calculus Midterm:
Recommended prep: 8-12 hours

Can use:

- assignment weight
 - course difficulty
 - time remaining
-

5. Smart Notifications

Examples:

You haven't started Assignment 3 and it's due in 4 days.

or:

You have 11 hours of estimated work this weekend.

6. Collaborative Planning

Group projects:

- shared deadlines
- teammate reminders
- shared task boards

Potentially huge demo value.

Recommended Architecture

Fastest Hackathon Architecture

Frontend

Best Choice

Next.js + Tailwind

Why:

- fast deployment
- easy UI
- good file upload support
- modern appearance quickly

UI Components

- shadcn/ui
 - Framer Motion
 - FullCalendar
-

Backend

Option 1 (Recommended)

Python FastAPI

Why:

- excellent PDF tooling
- great AI ecosystem
- easier OCR
- faster structured parsing

Option 2

Node.js Express

Better if team is stronger in JS.

AI Layer

Recommended

Gemini API

Why:

- multimodal
 - PDF understanding
 - likely sponsor alignment at GDG event
 - judges may appreciate ecosystem use
-

Database

Fastest

Supabase

Store:

- user uploads
- parsed tasks
- schedules
- reminders

Alternative:

- Firebase
 - PostgreSQL
-

Hosting

Easiest

- Vercel (frontend)
 - Railway/Render (backend)
-

Important Technical Challenge Research

Biggest Challenge: Date Extraction

Problem

Syllabi are inconsistent.

Examples:

```
Assignment 2 Due: October 11
```

vs

```
A2 submission deadline – 11 Oct.
```

vs

```
Week 5 deliverable (submit before midnight)
```

The parser must normalize all formats.

Solution Strategy

Use LLMs for semantic extraction

Instead of regex-only parsing.

Workflow:

1. Extract raw text from PDF
 2. Send chunks to Gemini
 3. Ask for structured JSON
 4. Validate schema
 5. Store clean events
-

Suggested Prompt Engineering

Extraction Prompt

You are an academic schedule parser.
Extract all assignments, quizzes, labs, exams, lecture topics, and deadlines.

Return valid JSON only.

Fields:

- course_name
 - event_type
 - title
 - due_date
 - weight
 - description
-

Workload Prompt

Analyze the student's semester schedule.
Identify overload periods.
Suggest workload balancing strategies.

Competitive Analysis

Existing Competitors

1. Notion Calendar

Weakness:

- manual input
- no syllabus intelligence

2. Motion

Weakness:

- expensive
- not student-focused

3. Google Calendar

Weakness:

- entirely manual

4. MyStudyLife

Weakness:

- limited AI
- weak automation

Your Advantage

You are specifically targeting:

- university students
- syllabus ingestion
- AI workload balancing
- automatic semester planning

That niche is very compelling.

What Judges Will Care About

1. Real Problem

This solves a real pain point.

2. AI Actually Adds Value

Not just: "We added a chatbot."

AI is used for:

- extraction
- planning
- prioritization
- forecasting

3. Strong Demo

Visual transformations are critical.

4. Technical Complexity

You can discuss:

- OCR
- NLP
- structured extraction
- prompt engineering
- scheduling systems
- calendar standards

How To Make It Look More Advanced Than It Is

1. Use Strong UI

A clean UI can double perceived quality.

Important:

- animations
- color-coded courses

- loading states
 - timeline visualizations
 - progress bars
-

2. Fake Very Little

Hackathons allow partial implementations.

You can:

- hardcode one advanced feature
- simulate smart recommendations
- use curated demo PDFs

As long as core functionality is real.

3. Prepare Perfect Demo Inputs

Do NOT demo random PDFs.

Prepare:

- clean syllabi
- overloaded weeks
- interesting outputs

This matters massively.

Potential Viral Features

"Academic Doom Week" Detector

Students LOVE this.

Example:

```
⚠ Doom Week Detected
Week 10 contains:
- 3 assignments
- 2 midterms
```


- 1 presentation
Estimated stress level: catastrophic

Very shareable.

GPA Risk Predictor

Potential future feature:

Skipping this assignment may reduce projected final grade to 71%.

AI Semester Roast

Funny feature:

Your November schedule looks like a boss fight.

Hackathon judges LOVE memorable humor.

Best MVP Scope For 24 Hours

DO THIS

Must-Haves

- PDF upload
- AI extraction
- unified calendar
- workload warnings
- todo generation
- export calendar

Nice-To-Haves

- login
- notifications
- chatbot
- collaborative features

Avoid

- complex auth
- mobile app
- browser extensions
- real LMS integration
- massive backend complexity

Stay focused.

Suggested Team Roles

Person 1

Frontend/UI

Person 2

Backend/API

Person 3

AI parsing + prompt engineering

Person 4

Calendar integration + polish/demo

Recommended Project Names

Best names:

- SyllabusSync AI
- Syllabiq
- CourseFlow
- SemesterOS
- DeadlineAI
- UniMap
- Schedulr
- StudyPilot
- TaskTerm

- ChronoClass

Best overall:

SemesterOS

Because it sounds like:

- startup-y
 - scalable
 - professional
-

Winning Demo Strategy

Start With Pain

"Every semester students get overwhelmed by massive syllabi and scattered deadlines."

Show Chaos

Open a giant syllabus PDF.

Show Magic Moment

Upload.

Then instantly show:

- clean calendar
- task board
- overload detection
- AI study plan

This is the key moment.

End With Vision

"Today we support syllabi. Tomorrow we can support full academic life management across every LMS platform."

Final Recommendation

This is one of the strongest hackathon ideas because:

- clear problem
- immediate relatability
- strong visuals
- meaningful AI use
- feasible MVP
- scalable startup potential
- easy judging narrative
- highly demoable

Most teams will build:

- generic AI chat apps
- random productivity tools
- unfinished social apps

This project feels like:

- a real product
- with real users
- solving a real issue
- using AI properly

That combination is exactly what performs well at hackathons.

Suggested Tech Stack Summary

Frontend

- Next.js
- Tailwind
- shadcn/ui
- FullCalendar

Backend

- FastAPI
- Python

AI

- Gemini API

Parsing

- pdfplumber
- PyMuPDF
- Tesseract OCR

Database

- Supabase

Deployment

- Vercel
 - Railway
-

Fastest Possible MVP Timeline

Hours 1–3

- setup frontend/backend
- implement upload

Hours 3–6

- PDF extraction
- Gemini parsing

Hours 6–10

- calendar UI
- event rendering

Hours 10–14

- workload analysis
- AI planner

Hours 14–18

- export functionality

- polishing

Hours 18–22

- demo preparation
- visuals
- animations

Hours 22–24

- presentation practice
- bug fixing
- deployment

Extra Judge Bait Features

Add One Tiny AI Agent

Example:

"What should I work on tonight?"

Even a basic implementation makes the app feel futuristic.

Add Semester Heatmap

Visualize stress intensity.

This is VERY demo friendly.

Add Priority Scores

Example:

Assignment Priority Score: 92/100

Judges love scoring systems.

Final Verdict

This is absolutely worth building at GDGHacks.

It has:

- strong product-market fit
- realistic implementation scope
- flashy demos
- genuine usefulness
- easy storytelling
- scalable startup potential

If executed cleanly, this can absolutely place highly at a student hackathon.